A

BRAC UNIVERSITY
Department of Computer Science Engineering
Mid-Term Examination, Spring 2025
CSE 350: Digital Electronics and Pulse Techniques

BRAC UNIVERSITY
Inspiring Excellence

Total Marks: **50 (10+20+20)**
Time: **75 Minutes**

- Total no. of questions are **four (4)**. **Question 2** is ***compulsory***.
  Answer **any two** from **Question 1, 3, 4**.
- Figure in bracket [ ] next to each question indicates marks for that question.
- **Unless otherwise stated**, use $\boldsymbol{V_\gamma(diode) = 0.6\ V}$, $\boldsymbol{V_D(conducting\ voltage\ of\ diode) = 0.7\ V}$, $\boldsymbol{V_\gamma(transistor) = 0.5\ V}$, $\boldsymbol{V_{BE}(forward\ active) = 0.7\ V}$, $\boldsymbol{V_{BE}(saturation) = 0.8\ V}$, $\boldsymbol{V_{CE}(saturation) = 0.2\ V}$, $\boldsymbol{\beta_F = 30}$, $\boldsymbol{V_{BC}(reverse\ active) = 0.7\ V}$ for all the questions.

Name:	ID:	Section:

Question 1 [20]

For the given circuit in Figure 1, assume $V_{OL} = 0.2V$, $V_{OH} = 4.5V$, $\beta_F = 30$, $V_{CE}(saturation) = 0.2\ V$, $V_{BE}(saturation) = 0.8\ V$, $V_\gamma(transistor) = 0.5V$

Driver and Loads are identical where $R_{in} = 5k\Omega$, $R_B = 100k\Omega$, and $R_C = 2k\Omega$, $V_{cc} = 5\ V$, $V_R = -5\ V$

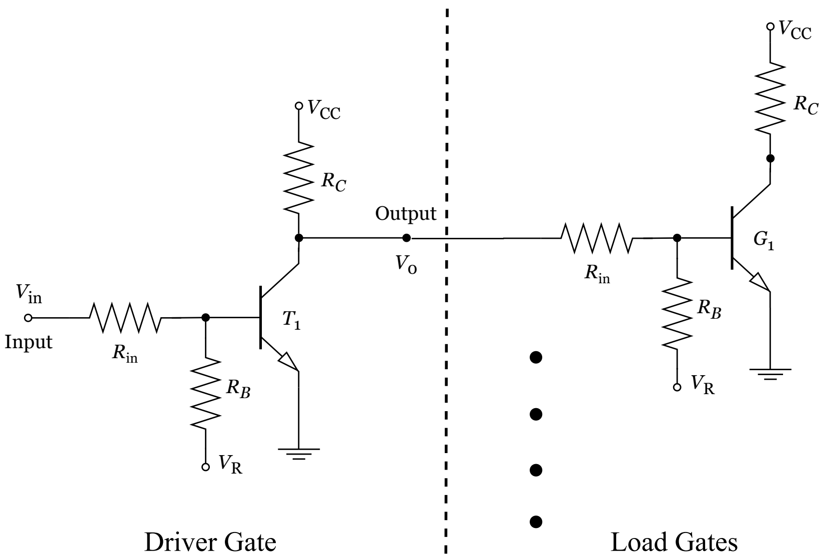

Figure 1

CO2	(a)	**Calculate** the power dissipation in the driver circuit, assuming **48 load gates** are connected, when V_{in} = **High**.	[10]
CO2	(b)	**Calculate** the noise margin of the driver circuit.	[10]

Question 2 [10]

CO1	(a)	**Implement** the logic function $Y = \overline{A} + \overline{B \cdot C}$, using **any combination of the logic families** you know. **Draw** the complete circuit. Then, mention the **operating mode** of your diodes/transistors when the **output is low**. [*You may simplify the function if possible* *Mentioning the values of circuit components is not necessary*]	[10]

Question 3 [20]

For the given circuit in Figure 2, assume $\boldsymbol{\beta_F = 30}$, $\boldsymbol{V_{CE}(saturation) = 0.1\ V}$, $\boldsymbol{V_{BE}(saturation) = 0.8\ V}$. Assume **Logical High** is $\mathbf{10}\boldsymbol{V}$ and **Logical Low** is $\mathbf{0.1}\boldsymbol{V}$.

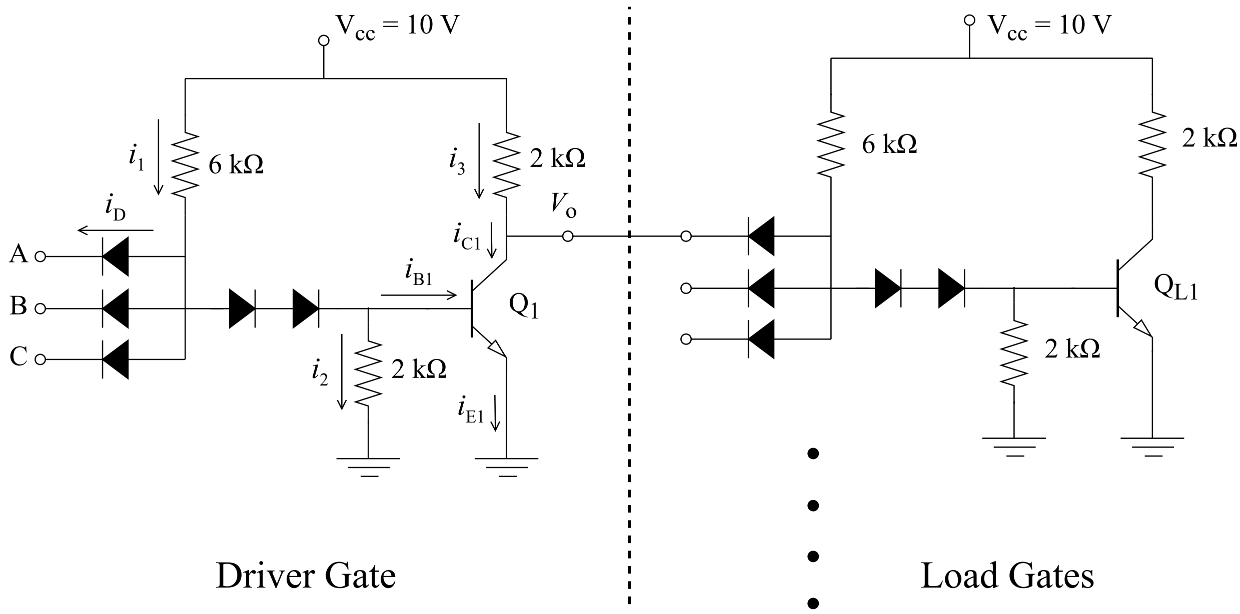

Figure 2

CO2	(a)	**Calculate** i_1, i_2, i_{B1}, i_{C1} and i_D, for A=Low, B=High, C=High in the driver circuit, **when load gates are not connected**. Then also find out the power dissipation (of the driver) for this case only.	[6]
CO2	(b)	**Repeat** Question-(a) for all input being logical high. **Prove** that Q1 transistor is in Saturation mode.	[6+1]
CO2	(c)	Given that the loads are now connected to the driver, **determine** the maximum fanout.	[7]

Question 4 [20]

For the given TTL circuit in Figure 3, $\boldsymbol{\beta_R = 0.2}$, $\boldsymbol{V_{BE(sat)} = 0.8V}$, $\boldsymbol{V_{CE(sat)} = 0.2V}$, and $\boldsymbol{V_{BC(reverse\ active)} = 0.7V}$. Assume **Logical High** is $\boldsymbol{6V}$ and **Logical Low** is $\boldsymbol{0.2V}$.

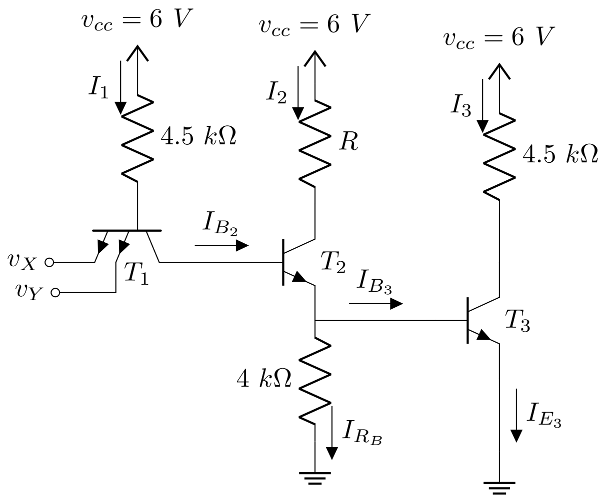

Figure 3

CO1	(a)	If $I_{E_3} = 4.74\ mA$ when both inputs are high, **calculate** the value of **R**.	[10]

For the circuit in Figure 4, assume **Logical High** is $\boldsymbol{5V}$ and **Logical Low** is $\boldsymbol{0V}$.

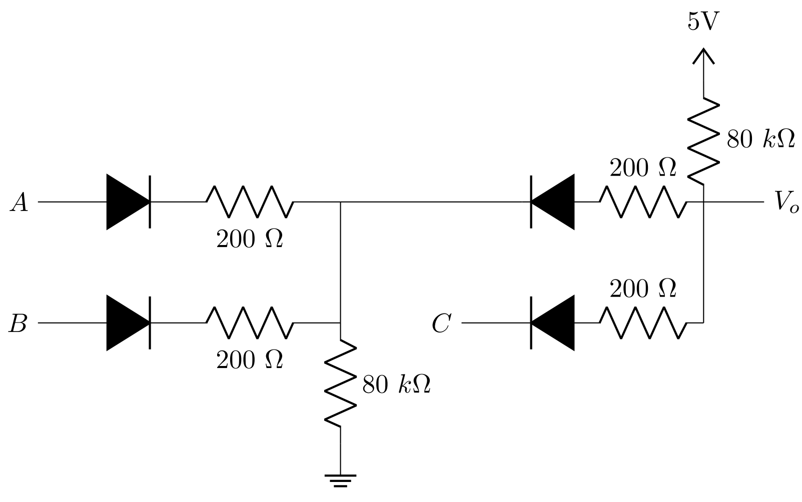

Figure 4

CO1	(b)	**Determine** the logic function this circuit is implementing.	[2]
CO1	(c)	**Calculate** the output voltage and power dissipation of the entire circuit when A=Low, B=High, C=Low.	[8]